TangibleData: Immersive Data Exploration with Touch

Ayush Bhardwaj
The University of Texas at Dallas
Richardson, TX, USA
ayush.bhardwaj@utdallas.edu

Jin Ryong Kim
The University of Texas at Dallas
Richardson, TX, USA
jin.kim@utdallas.edu

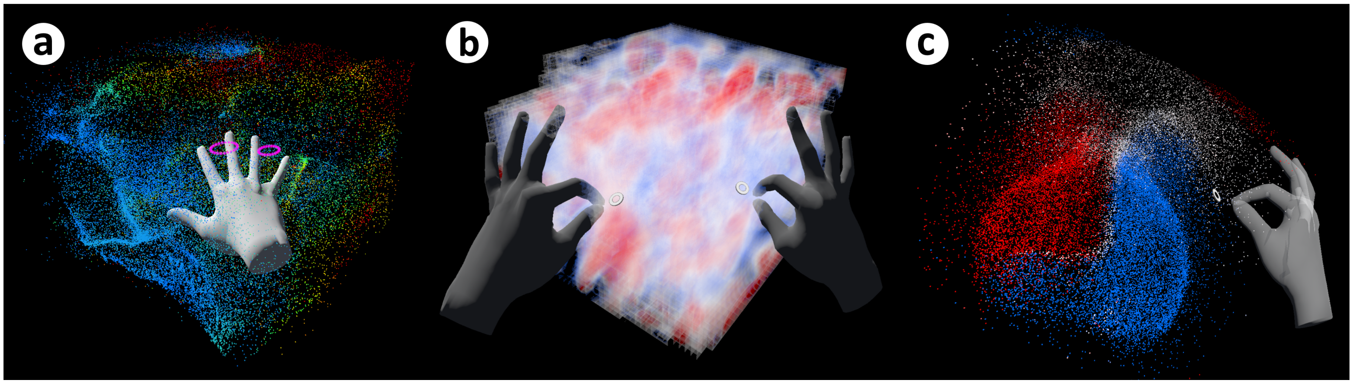

Figure 1: Hand interaction with fluid flow datasets while exploring different data properties. (a) Velocity direction using particle visualization (wind), (b) a user scaling the volume visualization of velocity data (wind), and (c) a user rotating particle visualization of the vorticity data (water).

ABSTRACT

We demonstrate an interactive 3D data visualization tool called TangibleData. TangibleData adapts hand gestures and mid-air haptics to provide 3D data exploration and interaction in VR using hand gestures and ultrasound mid-air haptic feedback. We showcase different types of 3D visualization datasets with different data encoding methods that convert data into visual and haptic representations for data interaction. We focus on an intuitive approach for each dataset, using mid-air haptics to improve the user's understanding.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; **Virtual reality**; **Visualization**; **Gestural input**.

KEYWORDS

Data visualization, haptics, immersive analytics, virtual reality

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1 INTRODUCTION

Data visualization has played a significant role in conveying visual representations of information to provide a greater understanding of trends and patterns in data. Visualizing massive amounts of data in 3D is becoming increasingly important so that users can explore multi-axis data interactively. Although 3D data visualization makes it possible to perceive depth, it is still important to effectively analyze large amounts of data from multiple perspectives and understand the spatial features of the dataset to provide a more in-depth understanding.

There are many cases where visual information alone is insufficient to present data due to its multidimensional nature. Challenges arise in accurately deciphering depth, spatial relationships, and intricate patterns solely through visual means. Solely relying on visuals results in problems like occlusion that leads to data misinterpretation and misunderstanding of data. Furthermore, as datasets grow in size and complexity, the burden on visual processing alone becomes increasingly overwhelming, necessitating alternative modes of data comprehension beyond visual modalities.

Our approach allows users to directly "touch and feel" datasets using hand-tracking and mid-air ultrasound haptic display through bare hands while interacting with data (see Figure 1 [Bhardwaj et al. 2021, 2022]). We integrated a number of dataset (fluid flow, 3D scatterplot, and medical) and transformed them into tangible data representations, allowing users to freely explore the data with touch sensation [Jansen et al. 2013, 2015]. Users can interact with data with bare hands and feel the implications of data to have a better understanding of data, while seeing multi-layered datasets.

Figure 2: User interaction. (a) A user with the hardware setup, (b) experiencing the vorticity of the wind data, and (c) a user interacting with human head dataset using real-time slicing feature

2 DEMO EXPERIENCE

At the SIGGRAPH Emerging Technologies, we will focus on demonstrating tangible user interactions with different datasets. The attendees will be asked to take a seat and wear a VR headset (see Figure 2(a)). After calibration is completed, they will then go through basic gestures to interact and manipulate the data. After a brief tutorial, they will use their hands to interact and feel the data with different data sets. The demo will last approximately 2-3 minutes per attendee. Detailed demo experience with each dataset is described as follows.

2.1 Fluid Flow

Users will experience two dynamic fluid flow datasets: wind and water. Each dataset uses three distinct visualization techniques: plane, volume, and particle visualization. Users will explore the following fluid properties: velocity magnitude in 3D space, recirculation regions, and fluid vorticity, through three specialized haptic rendering techniques. Each technique harnesses user's hand contact areas ("point of touch") to sample the fluid flow data. The velocity has been computed across all three axes (x, y, z) per timestep, transforming it into haptic feedback intensity (ranging from 1.64mN to 3.43mN) at a consistent frequency of 120 Hz (see Figure 1(a)). For the direction of velocity, we derived resultant vectors from all axes and translated them into haptic cues along the resultant vector's direction at frequencies from 80 Hz to 220 Hz, with a steady intensity of 3.43mN. For the vorticity representation, we employed clockwise and counterclockwise haptic rotations to convey vortex formation, varying rotational frequencies (from 60Hz to 100Hz), and intensities (ranging from 1.64mN to 3.43mN) to illustrate rotational strength (see Figure 2(b)). Through these interactive haptic techniques, users engage with fluid flow datasets, gaining tactile insights into the intricate dynamics of fluid behavior.

2.2 3D Scatterplot

Users will be engaged with a 3D scatterplot dataset featuring dense clusters, where the intricacies of cluster density come to life through mid-air haptic feedback. Our approach involves leveraging the frequency of collisions with the user's hand to modulate the strength of haptic sensations. To achieve this, we've engineered a density multiplier that compares the total number of data points within a cluster to those currently under user interaction. This dynamic multiplier informs real-time calculations of haptic intensity, ensuring a responsive and informative user experience. Normalizing the multiplier within the haptic intensity range enhances consistency and clarity, granting users a spatial advantage as they navigate and comprehend density variations across different clusters.

2.3 Medical Dataset

Users will delve into two distinct medical datasets: the human lung and the human head, with each offering unique insights. The human lung dataset elucidates the volume of cancerous tissues within the lungs. Users will interact with randomly spotted tissues and perceive volume information through haptic interaction. We utilized simple shapes like tetrahedrons within the volume mesh to estimate tissue volumes and transformed that into haptic intensity and frequency across the dataset. The human head dataset consists of multiple layers. We incorporated distance-to-tactile coding to present the proximity to the target location by varying the haptic feedback intensity based on the distance to the target. We also supported real-time slicing to allow users to explore multi-layered datasets without switching between different layers (see Figure 2(c)).

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